

Chapitre 1. Homography and Applications

1.1 Overview

1.2 Image Transformations

1.3 Homography

1.4 Computing the Homography

1.5 Dealing with Outliers: RANSAC

1.6 Image stitching

Chapitre 1. Homography and Applications

1.1 Overview

We suppose that we take a set of images of the scene from roughly the same viewpoint, but with rotating the camera.

As we are going to make sure that as we rotate the camera, the fields of views of the images overlap. So we want to take this set of images and automatically create a large image.



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1.1 Overview

We want to stitch them together to create a larger image or panorama.

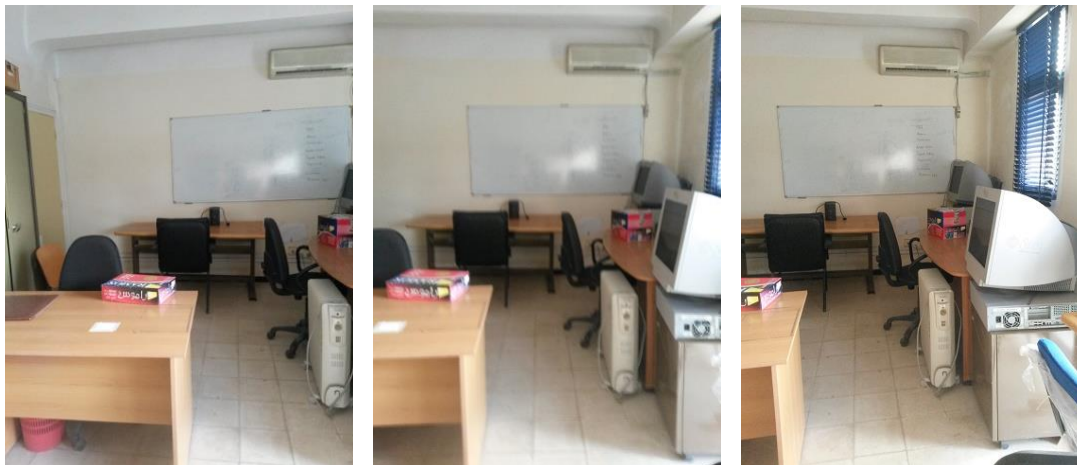


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1.1 Overview

How would we align these images?

Find corresponding points (using feature detectors like SIFT)

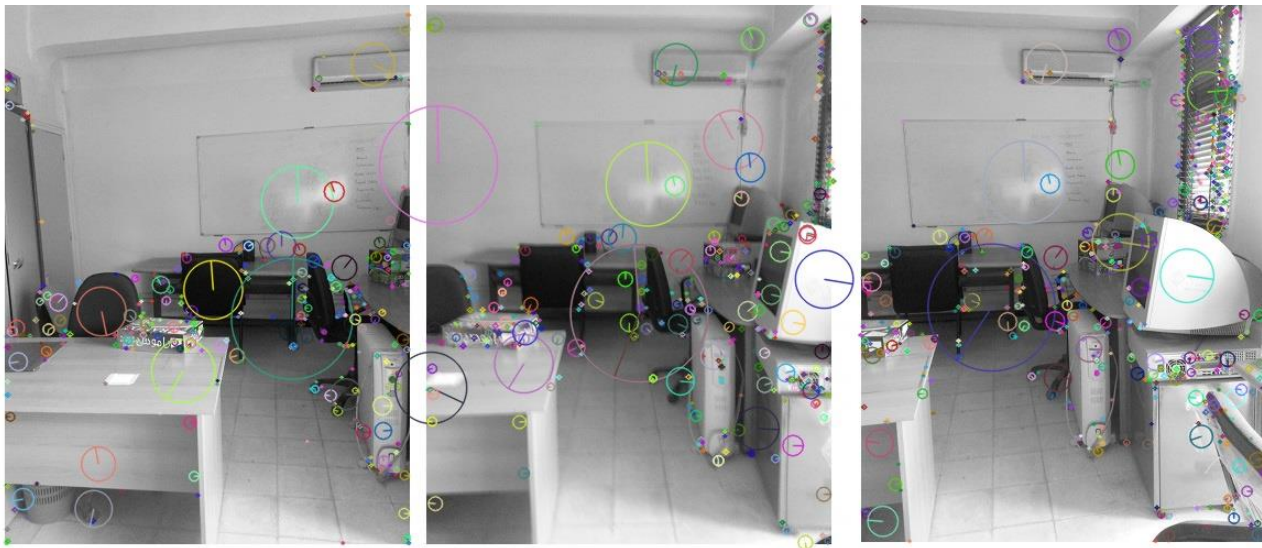


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1.1 Overview

How would we align these images?

Find corresponding points (using feature detectors like SIFT)



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1.1 Overview

Find geometric relationship between the images.

What is the transformation that take you from one image to another.

So we can take an image and wrap it to the coordinate frame of the other image.

This transformation is called the Homography.



Image 1



Image 2

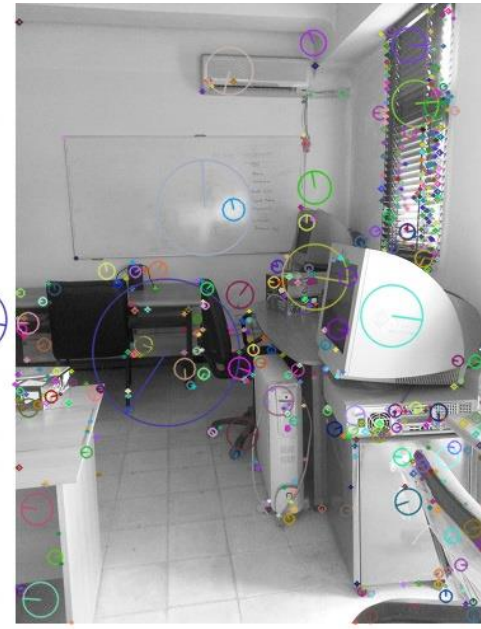


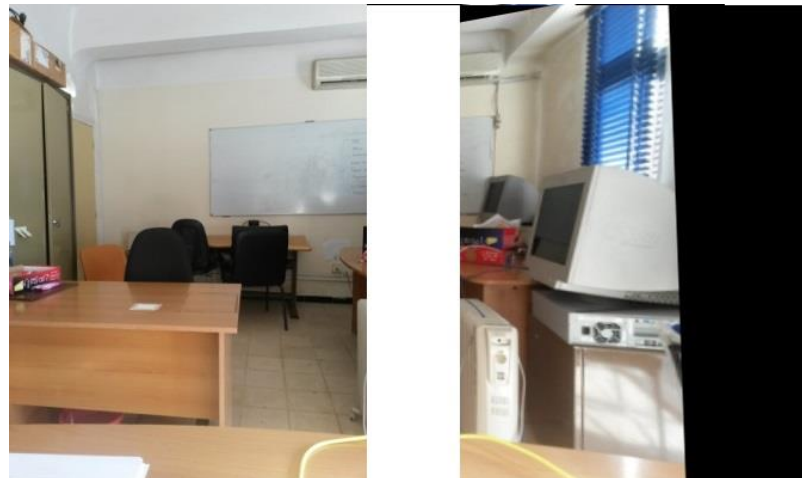
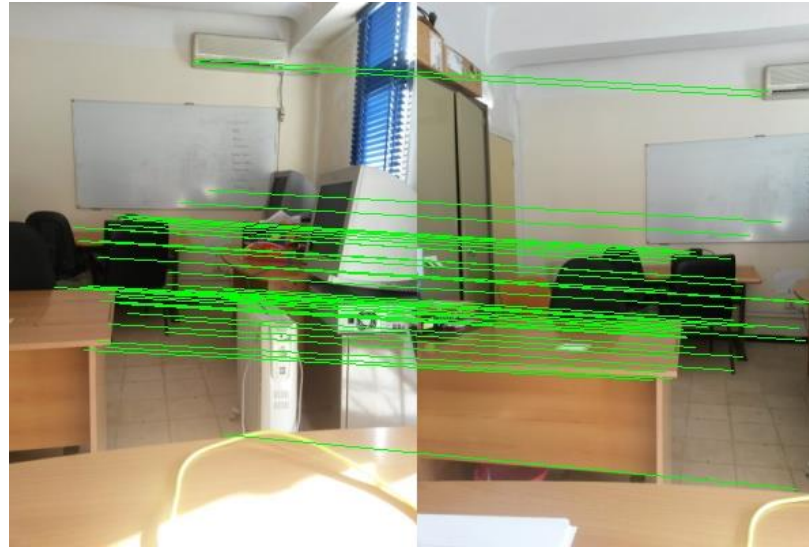
Image 3



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1.1 Overview

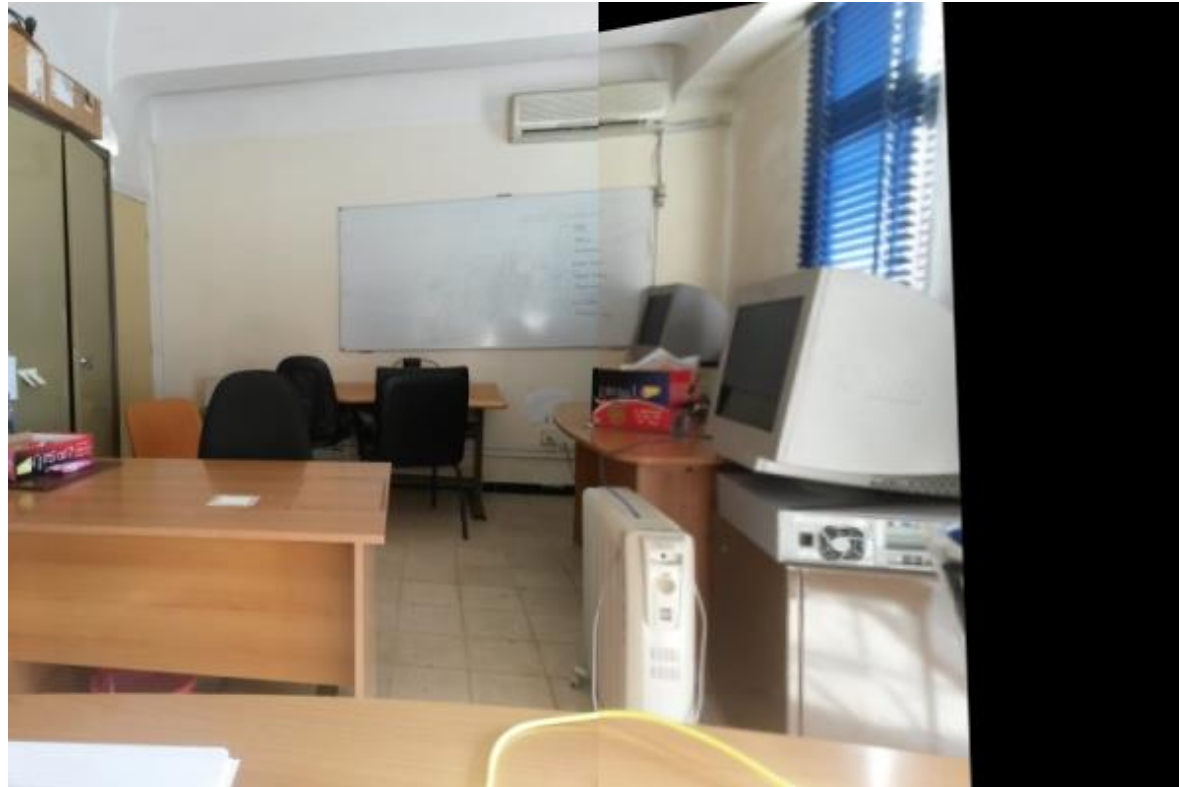
Example:
Matching Sift
features and
warping the left
image



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1.1 Overview

Example:
Panoramic image



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1.2 The 2D Image Transformations

Image Filtering: Change range(Brightness)

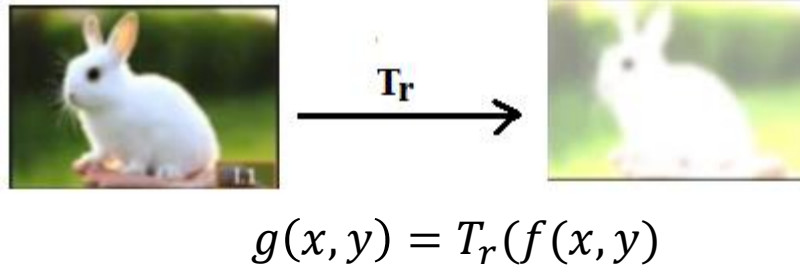


Image Warping: Change of Location



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1.2 The 2D Image Transformations

Rotation



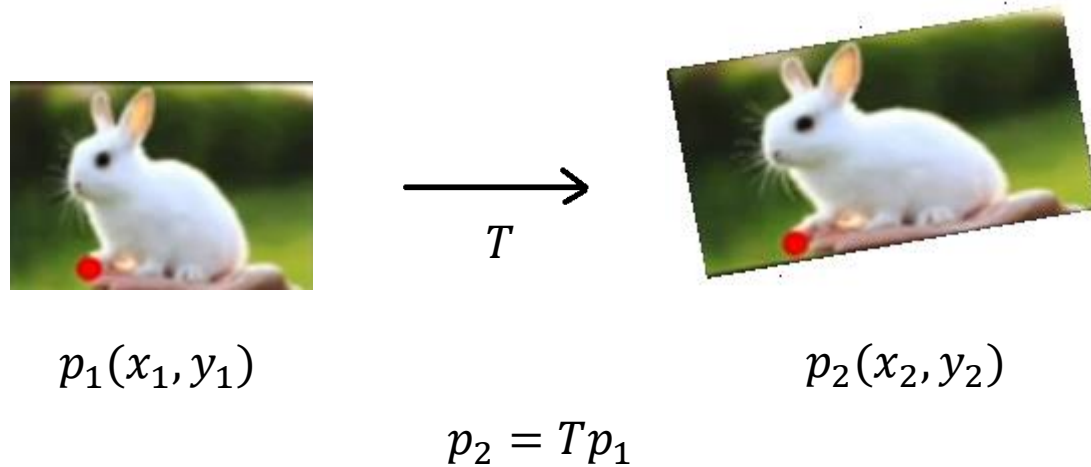
$$g(x, y) = f(T(x, y))$$

Projective



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1.2 The 2D Image Transformations



T can be represented by a Matrix

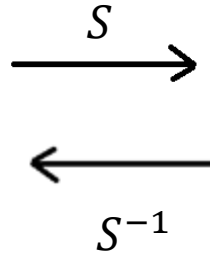
$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = T \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

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1.2 The 2D Image Transformations



$p_1(x_1, y_1)$



$p_2(x_2, y_2)$

$$p_2 = Tp_1$$

T can be represented by a Matrix (scaling)

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = S \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

$$S^{-1} = \begin{bmatrix} 1/a & 0 \\ 0 & 1/b \end{bmatrix}$$

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1.2 The 2D Image Transformations

2D Rotation

$$x_1 = r \cos(\varphi), y_1 = r \sin(\varphi)$$

$$x_2 = r \cos(\varphi + \theta)$$

$$x_2 = r \cos(\varphi) \cos(\theta) - \sin(\varphi) r \sin(\theta)$$

$$x_2 = x_1 \cos \theta - y_1 \sin(\theta)$$

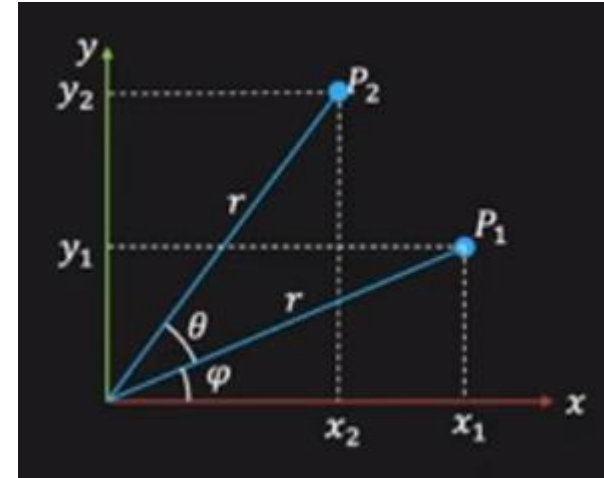
$$y_2 = r \sin(\varphi + \theta)$$

$$y_2 = r \cos(\varphi) \sin(\theta) + \sin(\varphi) r \cos(\theta)$$

$$y_2 = x_1 \sin \theta + y_1 \cos(\theta)$$

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = R \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

$$R^{-1} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$



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1.2 The 2D Image Transformations

Skew



Horizontal Skew: $x_2 = x_1 + m_x y_1$, $y_2 = y_1$

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = S_x \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 1 & m_x \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$



Vertical Skew: $x_2 = x_1$, $y_2 = y_1 + m_y x_1$

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = S_y \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ m_y & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$



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1.2 The 2D Image Transformations

Any other transformation

Any transformation of the form:

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = T \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

- Origin maps to the origin
- Lines map to lines
- Parallel lines remain parallel
- Closed under composition:

$$p_2 = T_{21}p_1, \quad p_3 = T_{32}p_2, \quad p_3 = T_{31}p_1$$

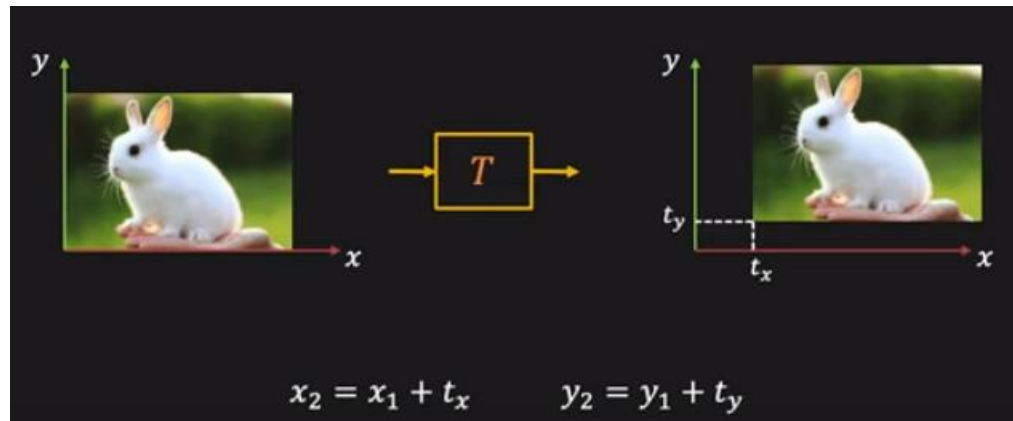
$$p_3 = T_{32}T_{21}p_1 \quad \Rightarrow \quad T_{32}T_{21}p_1 = T_{31}p_1$$

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1.2 The 2D Image Transformations

Homogeneous coordinates

This translation can't be expressed by a 2x2 matrix.



This limitation can be solved by bumping things up to a representation by one dimension.

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1.2 The 2D Image Transformations

Translation can't be expressed by a 2x2 matrix.

To express linearity, **Homogeneous coordinates** are widely used in science of engineering.

The homogeneous representation of a 2D point $p=(x,y)$ is a 3D point $\tilde{p} = (\tilde{x}, \tilde{y}, \tilde{z})$.

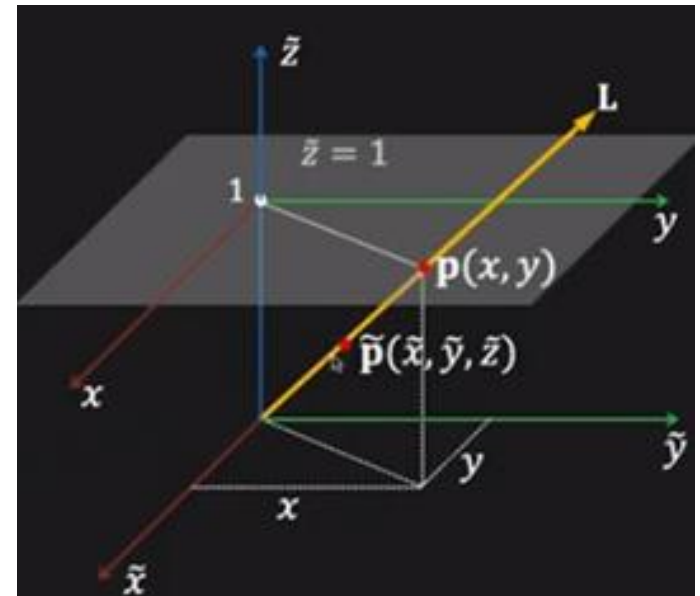
The third coordinate $\tilde{z}$ is fictitious such that:

$$x = \frac{\tilde{x}}{\tilde{z}}, \quad y = \frac{\tilde{y}}{\tilde{z}}$$

All points of the line (L) are equivalent to one another.

Every line on (L), except the origin, represents the homogeneous coordinates of $p(x,y)$

$$p \equiv \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \equiv \begin{bmatrix} \tilde{z}x \\ \tilde{z}y \\ \tilde{z} \end{bmatrix} \equiv \begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \end{bmatrix} = \tilde{p}$$



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1.2 The 2D Image Transformations

Homogeneous coordinates

Examples:

Scaling:

$$\begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \end{bmatrix} = \begin{bmatrix} sx & 0 & 0 \\ 0 & sy & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Translation:

$$\begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \end{bmatrix} = \begin{bmatrix} 1 & 0 & tx \\ 0 & 1 & ty \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

We can get a composition of these transformation (rotation+Scaling+Translation, ..) in a single transformation. We multiply these transformation and get a (3x3) transformation.

Affine transformation: Any transformation of the form:

$$\begin{bmatrix} x_2 \\ y_2 \\ 1 \end{bmatrix} \equiv \begin{bmatrix} \tilde{x}_2 \\ \tilde{y}_2 \\ \tilde{z}_2 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{y}_1 \\ \tilde{z}_1 \end{bmatrix}$$



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1.2 The 2D Image Transformations

Any affine transformation :

- Origin does not necessarily maps to the origin
- Lines map to lines
- Parallel lines remain parallel
- Closed under composition:

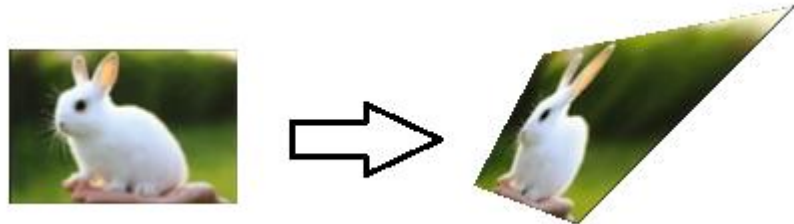
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1.3 Homography

Projective transformation, Homography :
Any transformation of the form:

$$\begin{bmatrix} \tilde{x}_2 \\ \tilde{y}_2 \\ \tilde{z}_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{y}_1 \\ \tilde{z}_1 \end{bmatrix}$$

$$\tilde{p}_2 = H\tilde{p}_1$$

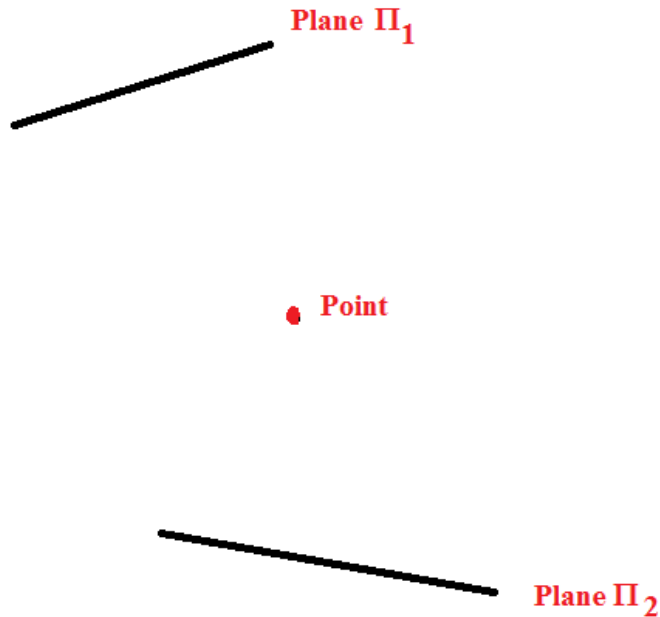


Is called Projective metrics: **HOMOGRAPHY**

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1.3 Homography

Projective transformation, Homography :
mapping one plane to another through a point:



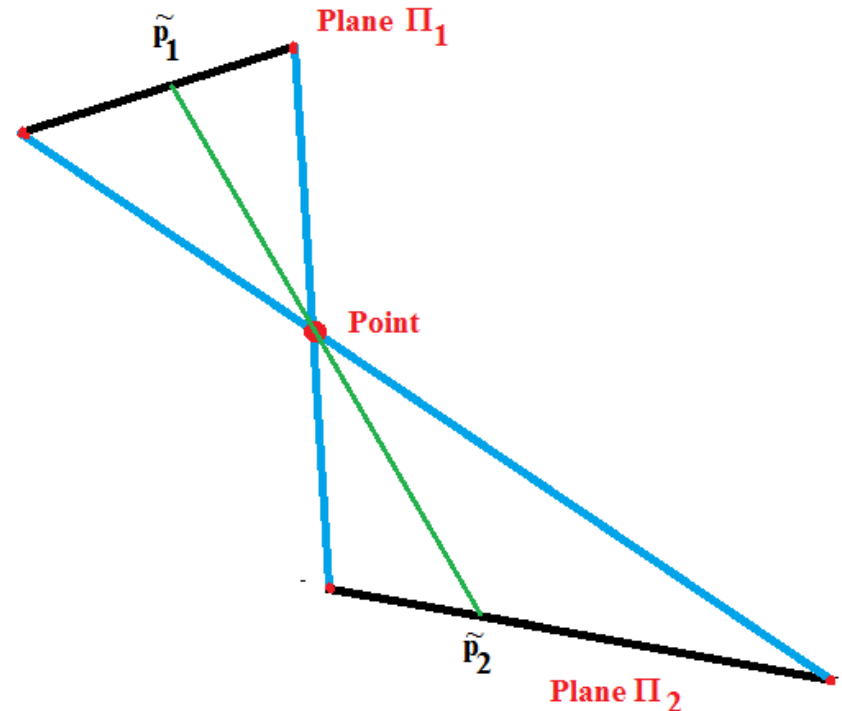
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1.3 Homography

Projective transformation, Homography :
mapping one plane to another through a point:

$$\begin{bmatrix} \tilde{x}_2 \\ \tilde{y}_2 \\ \tilde{z}_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{y}_1 \\ \tilde{z}_1 \end{bmatrix}$$

$$\tilde{p}_2 = H\tilde{p}_1$$



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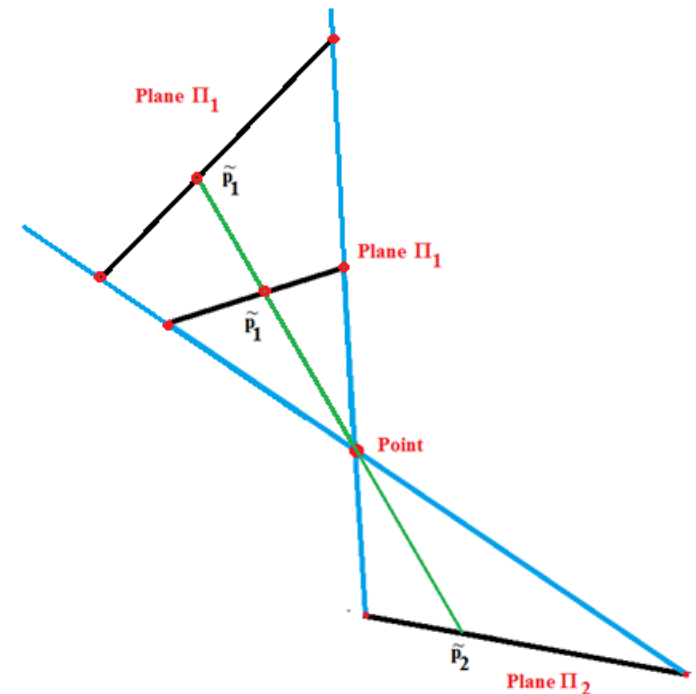
1.3 Homography

Projective transformation: Homography can only be defined up to scale.

Homography is the transformation matrix that take from one plane to another through a point of projection.

$$\begin{bmatrix} \tilde{x}_2 \\ \tilde{y}_2 \\ \tilde{z}_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{y}_1 \\ \tilde{z}_1 \end{bmatrix} \equiv k \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{y}_1 \\ \tilde{z}_1 \end{bmatrix}$$

Suppose that $y=Hx$. Then by elementary properties of matrix multiplication, $(kH)x=k(Hx)=ky$, so when $k \neq 0$, Hy and $(kH)y$ represent the same point.



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1.3 Homography

Projective transformation: Homography can only be defined up to scale. Homography is the transformation matrix that take from one plane to another through a point of projection.

$$\begin{bmatrix} \tilde{x}_2 \\ \tilde{y}_2 \\ \tilde{z}_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{y}_1 \\ \tilde{z}_1 \end{bmatrix} \equiv k \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{y}_1 \\ \tilde{z}_1 \end{bmatrix}$$

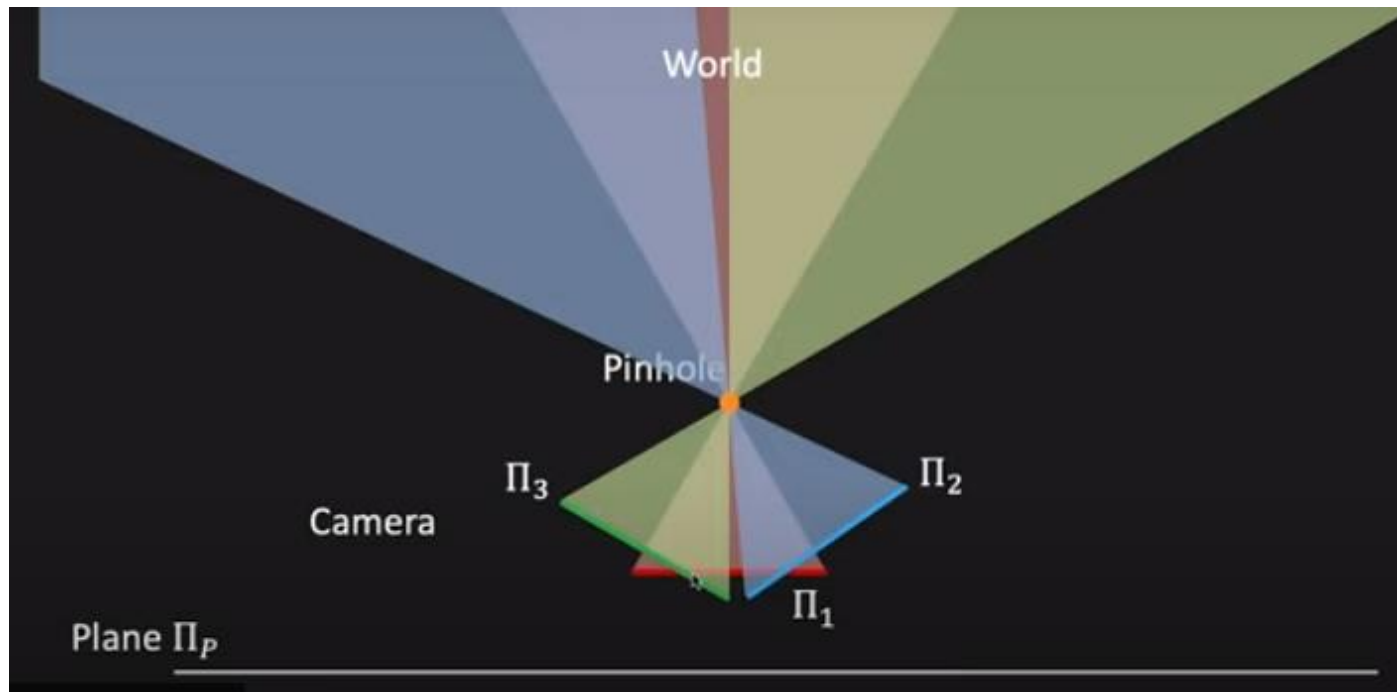


If we fix $\sqrt{\sum_{ij}(h_{ij})^2} = 1$, scale such that then 8 free parameters.

- Origin does not necessarily maps to the origin
- Lines map to lines
- Parallel lines does not necessarily remain parallel
- Closed under composition:

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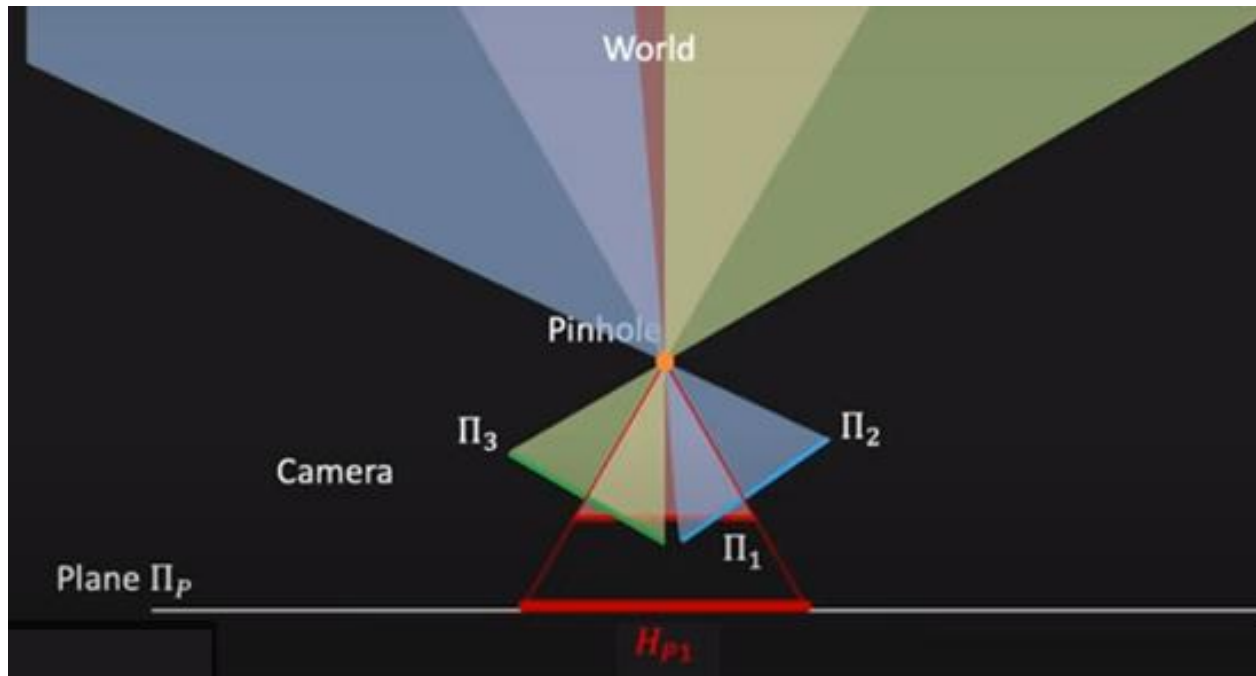
1.3 Homography



The plane Π_p share with Π_1 the same central projection. We can map Π_1 to Π_p using the Homography H_{1p}

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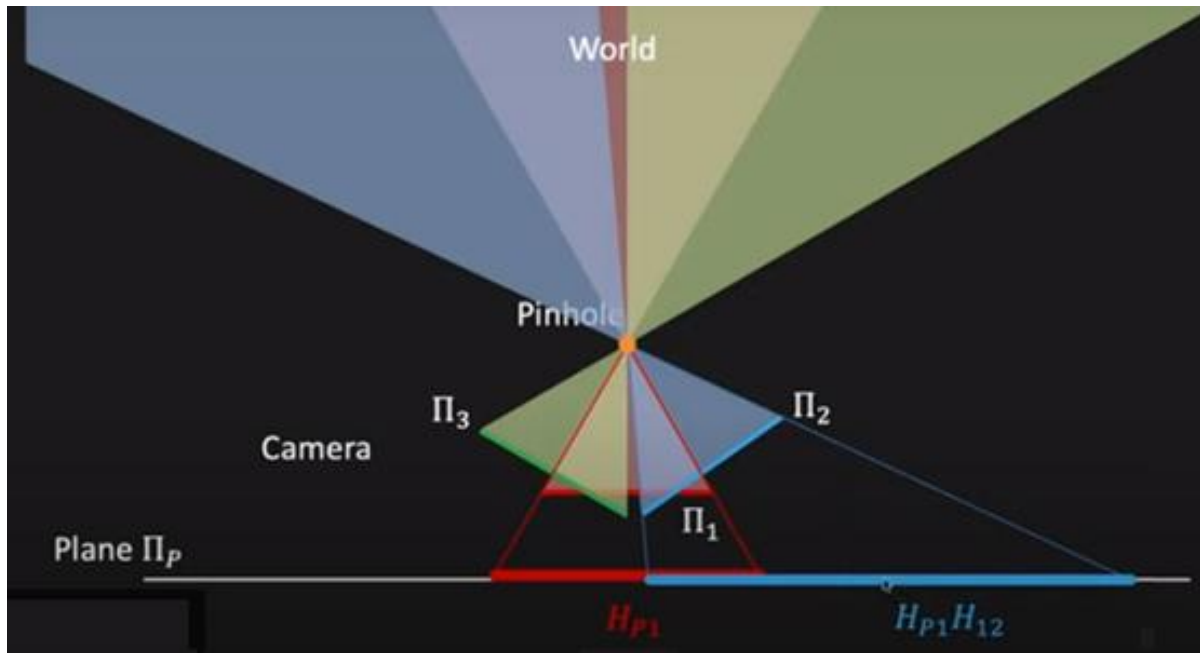
1.3 Homography



The plane Π_p share with Π_1 the same central projection. We can map Π_1 to Π_p using the Homography H_{1p}

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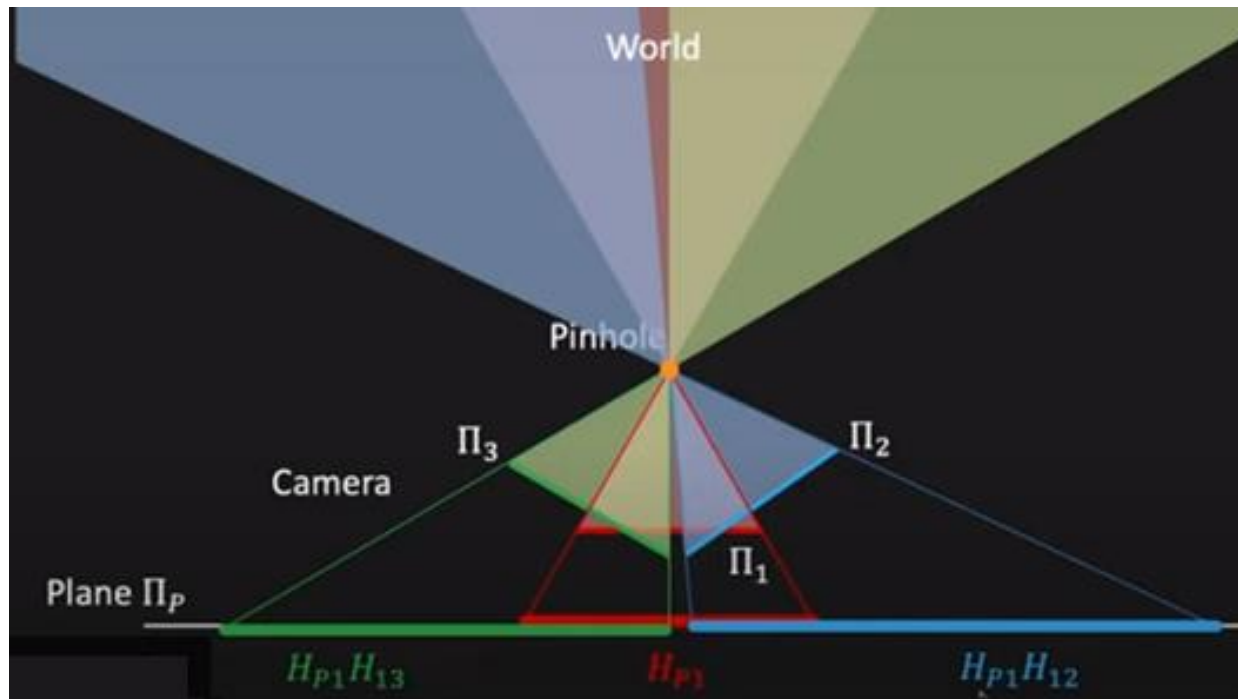
1.3 Homography



The plane Π_2 is mapped to Π_1 using the Homography H_{12} .

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1.3 Homography



We can map all images to a single plane by simply computing the Homographies between the images. This is useful for stitching panoramas.

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1.4 Computing Homography



Find the Homography that best agrees with the matches (SIFT descriptors).

Hypothesis:

- the images are acquired from the same view point.
- Or the scene points should lie on a same plane.
- Or the scene is really far away (scene is a plane at infinity).

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1.4 Computing Homography



Source

Destination

$$\begin{bmatrix} x_d \\ y_d \\ 1 \end{bmatrix} \equiv \begin{bmatrix} \tilde{x}_d \\ \tilde{y}_d \\ \tilde{z}_d \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$

9 unknowns, 8 degrees of freedom.

Then, at the minimum, 04 pairs of matching points

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1.4 Computing Homography

For a given Source pair i of corresponding points:

$$\begin{bmatrix} x_d \\ y_d \\ 1 \end{bmatrix} \equiv \begin{bmatrix} \tilde{x}_d \\ \tilde{y}_d \\ \tilde{z}_d \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix}$$

$$x_d = \frac{\tilde{x}_d}{\tilde{z}_d} = \frac{h_{11} x_s + h_{12} y_s + h_{13}}{h_{31} x_s + h_{32} y_s + h_{33}} = \quad y_d = \frac{\tilde{y}_d}{\tilde{z}_d} = \frac{h_{21} x_s + h_{22} y_s + h_{23}}{h_{31} x_s + h_{32} y_s + h_{33}} =$$



$$x_d(h_{31} x_s + h_{32} y_s + h_{33}) = h_{11} x_s + h_{12} y_s + h_{13}$$

$$y_d(h_{31} x_s + h_{32} y_s + h_{33}) = h_{21} x_s + h_{22} y_s + h_{23}$$

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1.4 Computing Homography

For a given Source pair i of corresponding points, we obtain a linear equation:

$$x_d(h_{31} x_s + h_{32} y_s + h_{33}) = h_{11} x_s + h_{12} y_s + h_{13}$$

$$y_d(h_{31} x_s + h_{32} y_s + h_{33}) = h_{21} x_s + h_{22} y_s + h_{23}$$

$$\Rightarrow \begin{bmatrix} x_s & y_s & 1 & 0 & 0 & 0 & -x_d x_s & -x_d y_s & -x_d \\ 0 & 0 & 0 & x_s & y_s & 1 & -y_d x_s & -y_d y_s & -y_d \\ \dots & \text{second pair of matched points} & & & & & & & \\ & \text{Third pair of matched points} & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \\ & & & & & & & & \end{bmatrix} \begin{bmatrix} h_{11} \\ h_{12} \\ h_{13} \\ h_{21} \\ h_{22} \\ h_{23} \\ h_{31} \\ h_{32} \\ h_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \cdot \\ \cdot \\ 0 \end{bmatrix}$$

A: Known

h: Unknown

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1.4 Computing Homography

Solve for h :

$$Ah = 0, \text{ such that } \|h\|^2 = 1$$

We define least squares problem:

$$\min_h \|Ah\|^2, \text{ such that } \|h\|^2 = 1$$

We know:

$$\|Ah\|^2 = (Ah)^T Ah = h^T A^T Ah, \text{ such that } \|h\|^2 = h^T h = 1$$

Solve for h :

$$\min_h \|h^T A^T Ah\|^2, \text{ such that } h^T h = 1.$$

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1.4 Computing Homography

We define the loss function $L(h, \lambda) = h^T A^T A h - \lambda(h^T h - 1)$

We derive L relatively to h :

$$2A^T A h - 2\lambda h = 0$$

$$A^T A h - \lambda h = 0 \quad \text{Eigen Values problem}$$

Solution: We choose the eigen vector h with smallest value of λ of $A^T A$ which minimize the loss function L

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1.5 Dealing with Outliers: RANSAC

Problem of Outliers:

We need to robustly compute transformation in presence of wrong matches.

If the number of outliers is $< 50\%$, then **RANSAC** (**R**ANdom **S**Ample **C**onsensus) to the rescue.

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1.5 Dealing with Outliers: RANSAC

General RANSAC Algorithm

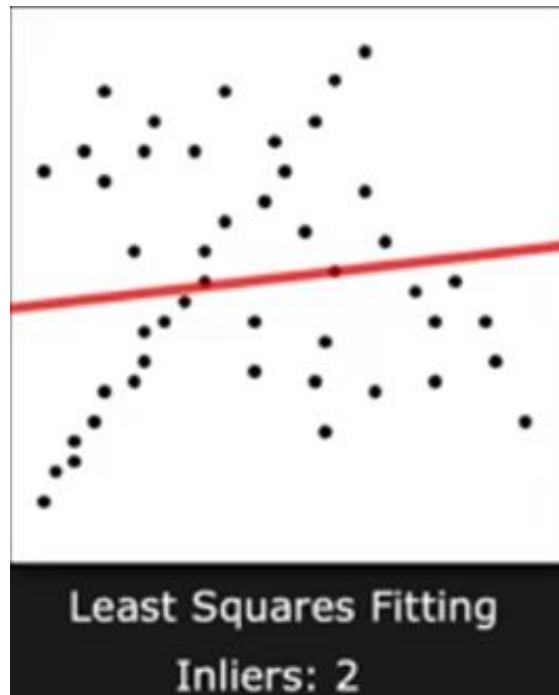
- 1- Randomly choose s samples. Typically, s is the minimum samples to fit a model.
- 2- Fit the model to the randomly chosen samples
- 3- Count the number M of data points (inliers) that fit the model within a measure of error ϵ .
- 4- Repeat 1..3 N times
- 5- Choose a model that has the largest number M of inliers.

For Homography, $N=4$, ϵ : acceptable alignment error in pixels.

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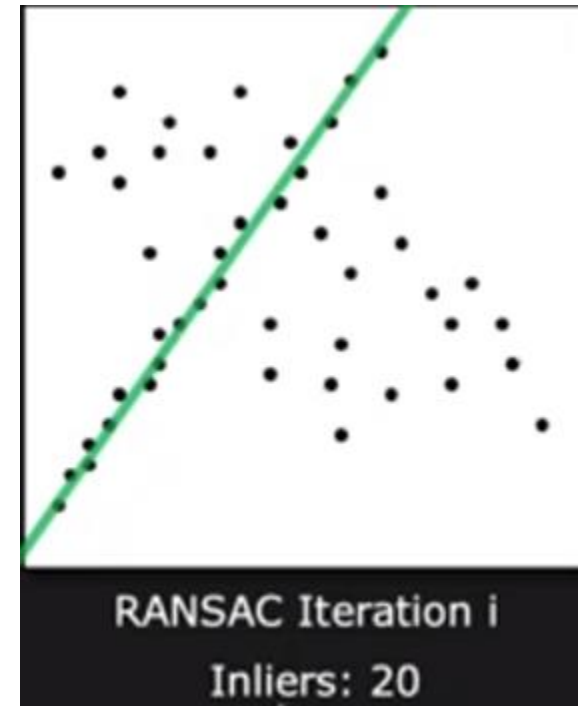
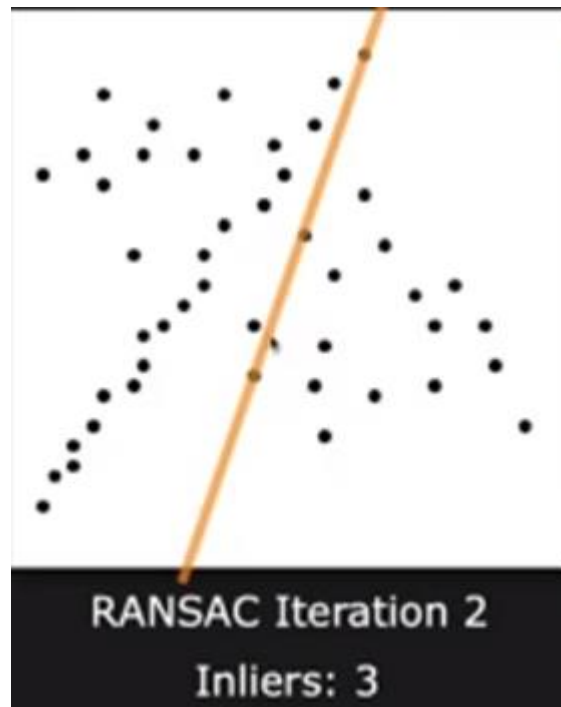
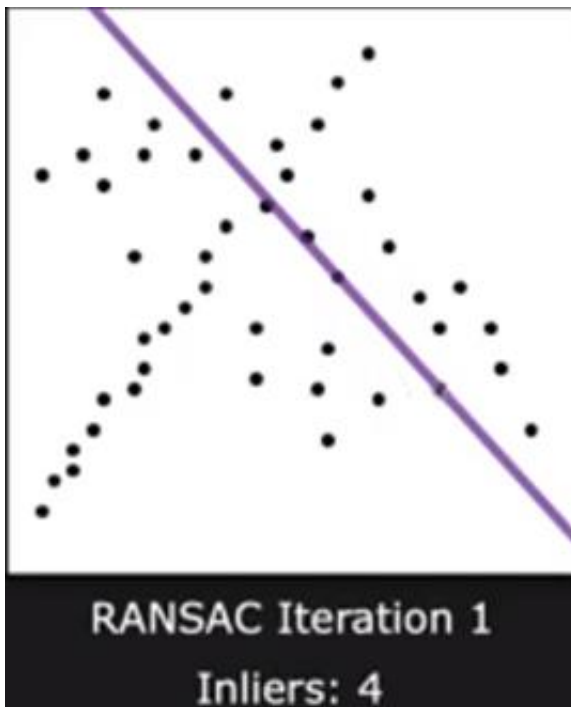
1.5 Dealing with Outliers: RANSAC

Example: Robust line fitting



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1.5 Dealing with Outliers: RANSAC



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1.6 Image Stitching

Given a transformation T , and an image $f(x,y)$,
we compute the transformed image $g(x,y)$



$f(x,y)$



$g(x,y)=T(f(x,y))$

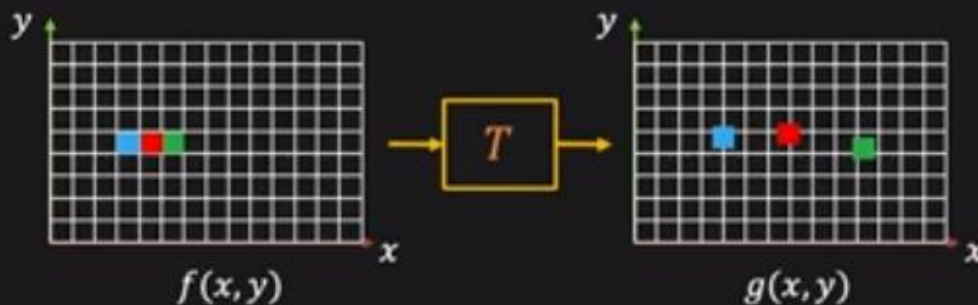
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1.6 Image Stitching

Forward Warping

Send each pixel (x, y) in $f(x, y)$ to its corresponding location $T(x, y)$ in $g(x, y)$

$$g(x, y) = f(T(x, y))$$



What if pixel lands in between pixels?
What if not all pixels in $g(x, y)$ are filled?

Can result in holes!

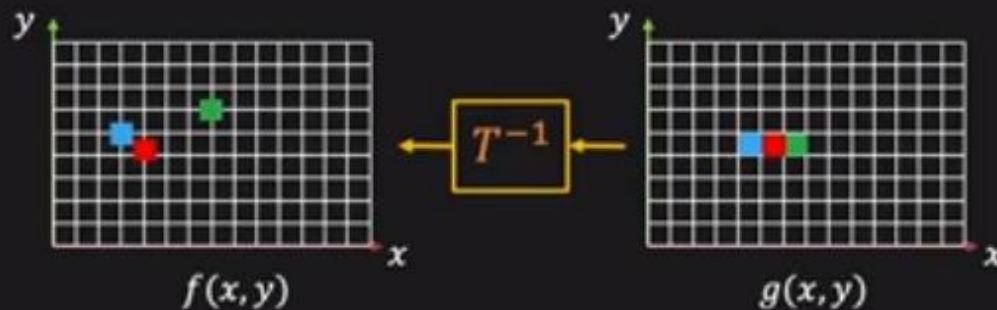
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1.6 Image Stitching

Backward Warping

Get each pixel (x, y) in $g(x, y)$ from its corresponding location $T^{-1}(x, y)$ in $f(x, y)$

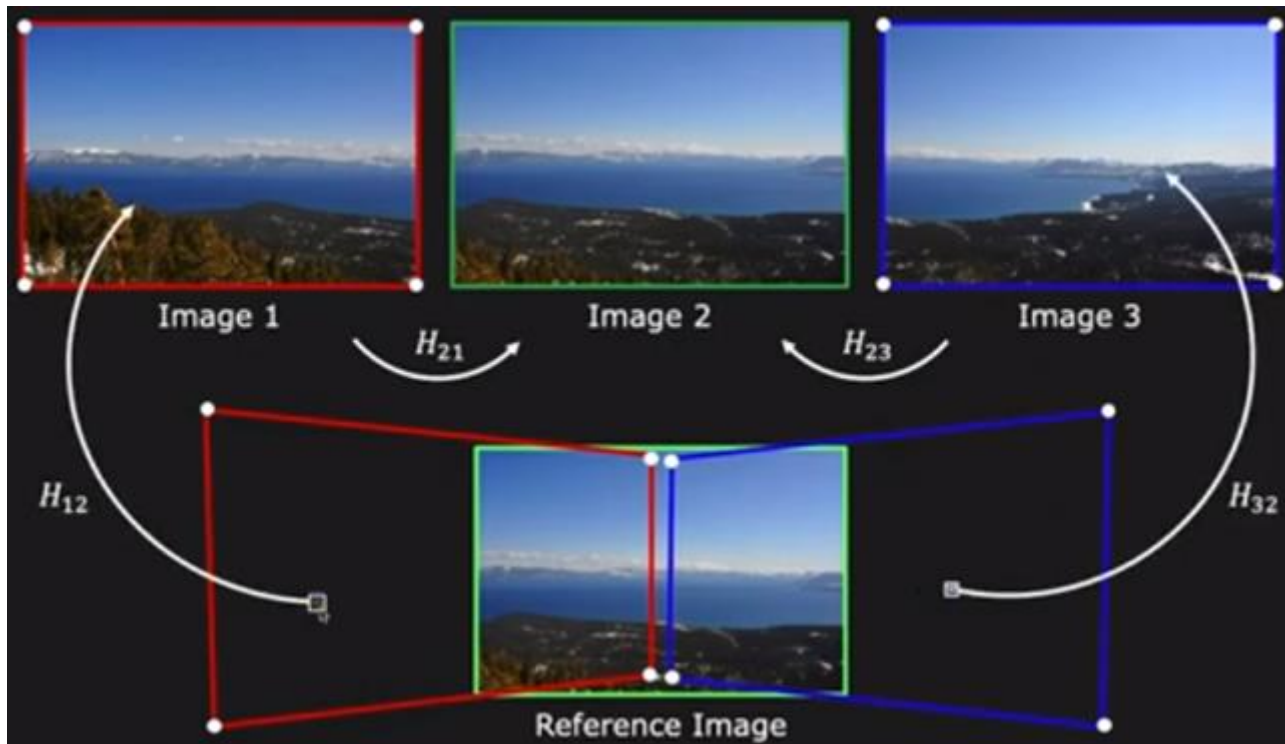
$$g(x, y) = f(T(x, y))$$



What if pixel lands between pixels?
Use **Nearest Neighbor** or **Interpolate**

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1.6 Image Stitching



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1.6 Image Stitching



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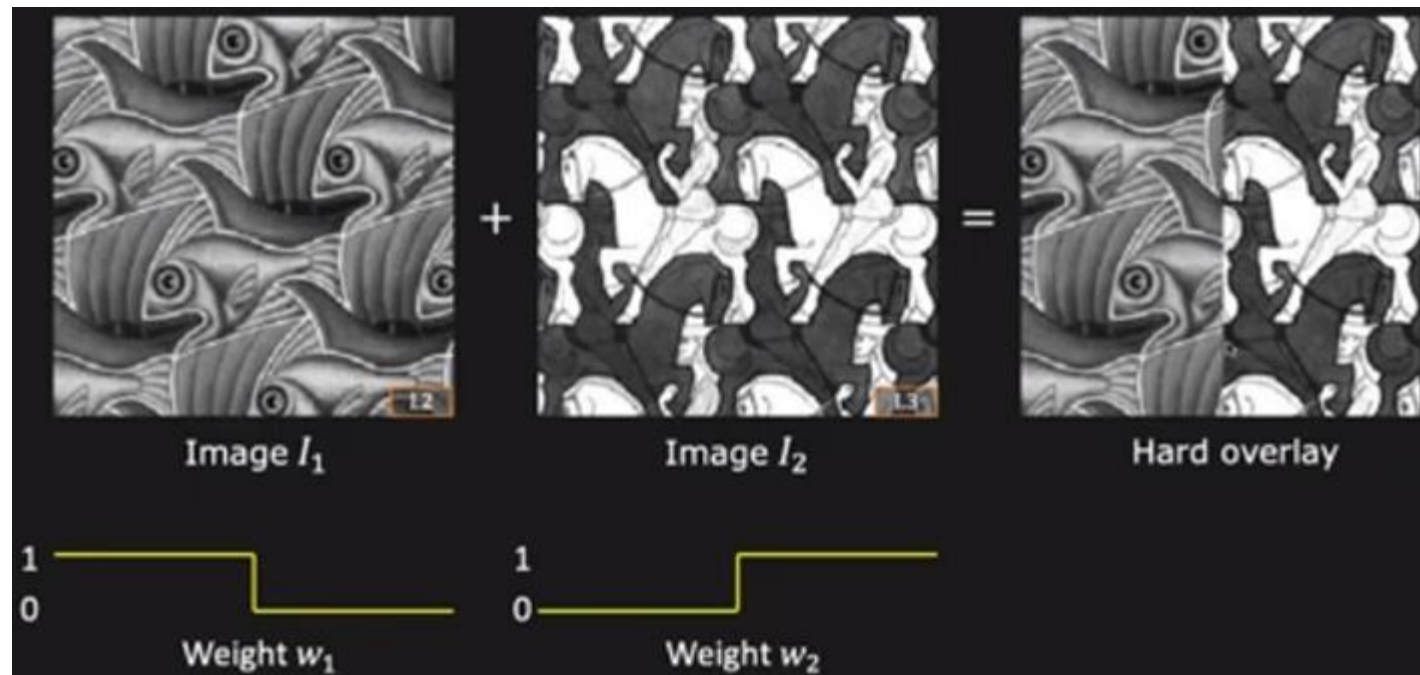
1.6 Image Stitching



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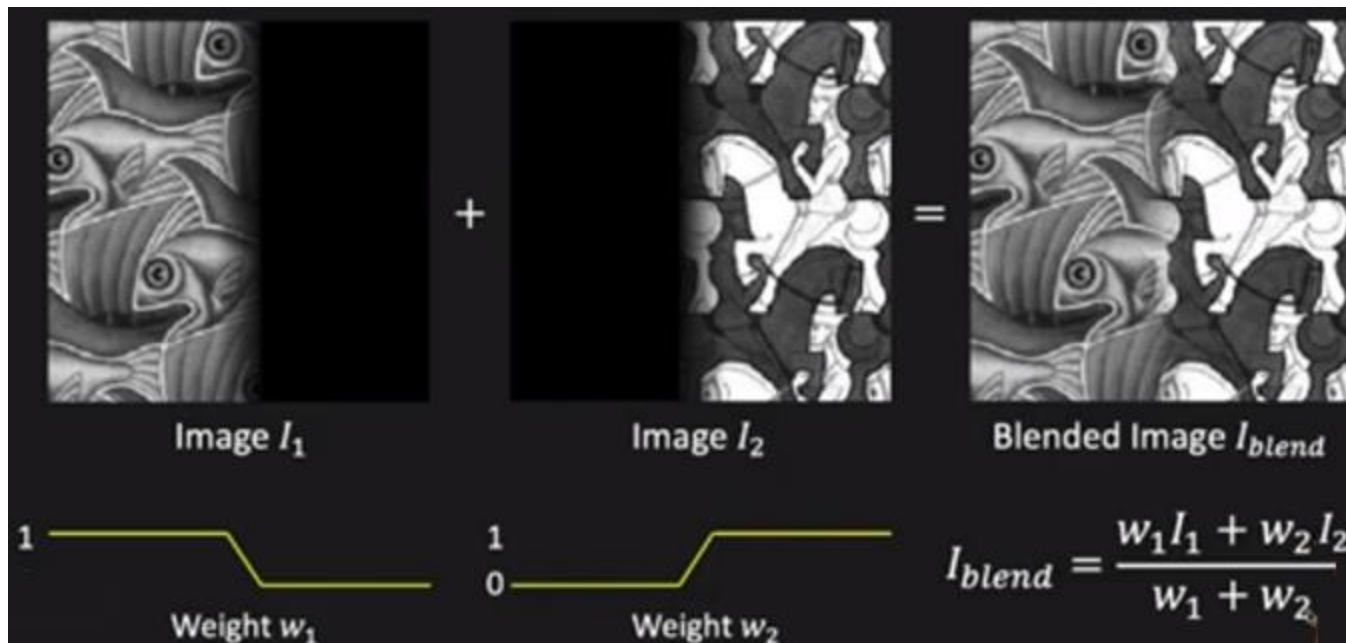
1.6 Image Stitching

We want to blend images I_1 and I_2 at the center.



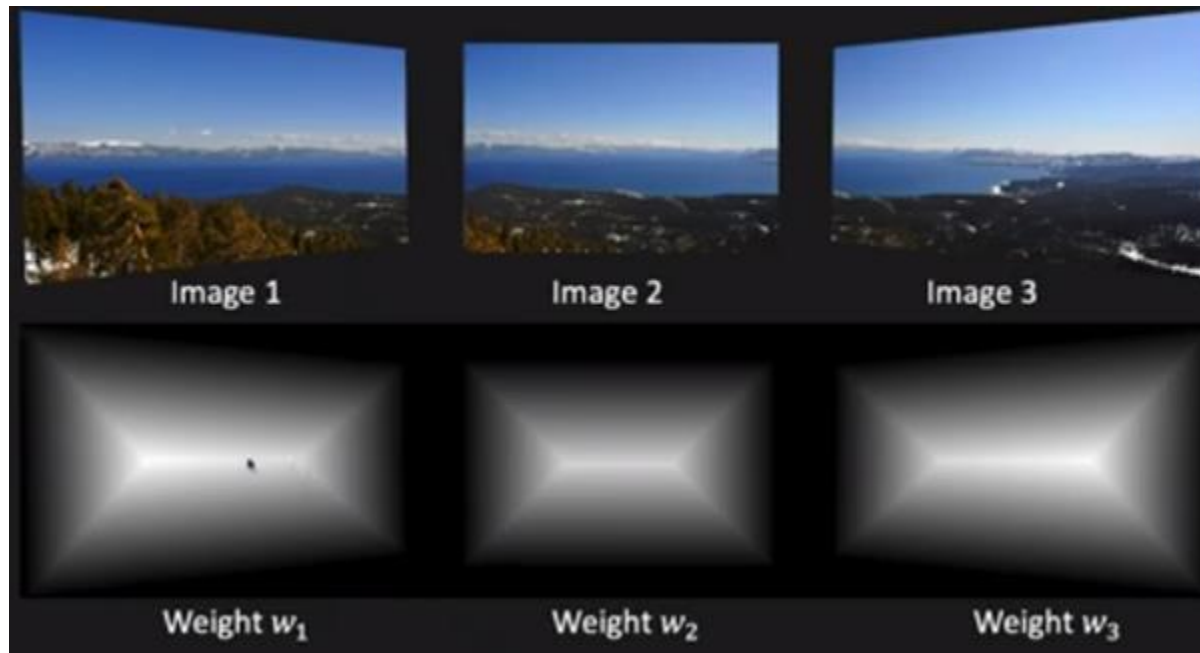
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1.6 Image Stitching



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1.6 Image Stitching



Pixels closer to the edges get a lower weight.

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1.6 Image Stitching

Weighted blending



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1.6 Image Stitching

